Photocatalytic degradation of microplastics: TiO_2-based catalysts Review

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Abstract Microplastics (size <5 mm) have emerged as a critical environmental pollutant, accumulating in aquatic ecosystems and threatening marine organisms and human health through trophic transfer. **Photocatalytic degradation** using **TiO_2-based catalysts** offers a sustainable solution, demonstrating significant potential in mitigating microplastic pollution. This review evaluates seven widely used photocatalysts under varying environmental conditions (e.g., pH), revealing diverse degradation efficiencies (4–100% mass loss) dependent on polymer type and irradiation parameters. Key findings highlight the role of electron-hole pair generation, redox reactions, and pH-dependent surface charge dynamics in optimizing catalytic performance. Acidic conditions (pH 3.0–4.5) enhance efficiency by reducing catalyst aggregation through protonation ($\equiv TiOH_2^+$), while neutral/alkaline pH favors less reactive species like H_2O_2. Notably, Ag/TiO_2 achieves 100% polyethylene degradation in 6 hours under UV irradiation at pH 3.0, outperforming other catalysts. Challenges such as prolonged irradiation requirements and catalyst instability under acidic conditions (12% TiO_2 leaching in 48 hours) are discussed, alongside recommendations for advancing industrial applications through hybrid material design and pH adaptability optimization.

Keywords: microplastics, photocatalytic degradation, TiO_2-based catalysts, environmental impact, redox reactions

1 Introduction

Microplastics (size <5 mm) are an important source of global environmental pollution and will continue to accumulate in the environment, especially in the aquatic environment. [1] First, due to its low weight, microplastics is easy to be transported by winds and flows, absorbing pollutants along the journey to the marine environment.[2, 3] Second, long-term existence of microplastics in the marine environment has harmful impacts on the health of marine organisms[4] , and subsequently humans through the trophic chain.[5] Compared to conventional filtration methods that merely transfer pollutants from water to solid waste, photocatalytic degradation offers a sustainable solution by mineralizing microplastics into CO_2 and H_2O.[1] However, challenges such as low solar utilization efficiency (<5%) and catalyst deactivation in complex matrices remain unresolved.[3] Some bare photocatalysts have been demonstrated to show good microplastics degradation efficiency.[6, 7] Among the detected photocatalysts, TiO_2 is extensively applied in wastewater microplastics processing industry. According to rough statistics, seven catalysts are used widely for unique advantages in processing different substances under certain circumstances, such as PH and temperature. This review systematically evaluates TiO_2-based photocatalysts' performance under varying environmental conditions, focusing on pH and polymer type interactions.

2 Photocatalytic degradation of microplastics

The degradation of (micro)plastics is a complicated process. Photocatalytic degradation needs to satisfy a variety of conditions to pursue the high efficiency.[8] TiO_2-based photocatalysts have been confirmed to be effective over numerous categories of plastics or chlorine-free polymers. Table 1 displays the results. Besides, the size of the polymer also matter. Suitable size of the catalyst are proved to be more potential to promote the process of redox subreaction with the experiment.(Table 2) Moreover, the photocatalysts accelerate the reactions only under the suitable environment. The pH, representing the concentration of H^+, contributes the most to the catalyst activity.

2.1 Mechanisms of the Photocatalysts

The target of a photocatalyst is to accelerate redox reactions under the condition of light shining. Generally, the photocatalyst reaction is composed of three critical subreactions(Figure 2). First, with an energy larger or equal to the energy threshold(E_g), the light irradiation activates the photocatalyst and creates electron-hole pairs. Second, electrons (e^-) are created in the conduction bands, while holes (h^+) are generated in the valence bands correspondingly. Third, the charge carriers induced by the light transfer to the surface of the photocatalyst, subsequently facilitate the process of redox reactions. [9, 10] The table concludes, lists and compares different properties of photocatalytic degradation of microplastics. [11]

Table 1: Performance comparison among different photocatalysts used in microplastics degradation.

Photocatalyst	Microplastics	Irradiation source	Irradiation time (hour)	Degradation efficiency (% mass loss)	Ref.
TiO_2	PP film	Xenon	144	4-8	[9] [12]
TiO_2	PS	UV	150	22.5	[10] [13]
TiO_2/Fe(St)$_3$	PS	UV	480	63.08	[11] [14]
Ag/TiO_2	PE	UV	6	100	[12] [15]
$FePcCl_{16}$-TiO_2	PVC film	UV	360	81	[13] [16]
Graphite-TiO_2	PVC	UV	30	17.24	[14] [17]

Notably in Table 1, the degradation efficiency of Ag/TiO_2 actually reach 100% within 6 hours under pH 3.0 UV irradiation, outperforming other catalysts by orders of magnitude. [14, 15] This efficiency surpasses TiO_2/Fe(St)$_3$ (63.08% in 480 hours) and $FePcCl_{16}-TiO_2$ (81% in 360 hours), highlighting the synergistic effect of Ag nanoparticle plasmon resonance and TiO_2 redox activity. According to the Maulana et la (2021), the degradation speed of the microplastics is mainly based on the particle size. This event is related to the ability of the microplastic to mix evenly in a solution. The article is not going to discuss about the mechanism back the abnormal situation. From the degradation efficiency and the comparison among the catalysts,[11] TiO_2 stands out in microplastics processing industry due to the high efficiency and the common condition of pH, which is about 3.0. Comparative analysis reveals that particle size further modulates this efficiency: 100–125 µm PE particles achieve complete degradation 20% faster than 150–250 µm particles under identical conditions (Table 2). This size-dependent trend aligns with the higher surface-area-to-volume ratio accelerating ROS diffusion. [17] .

Particle size ()	Percent Degradation (%)				
	0 min	30 min	60 min	90 min	120 min
100–125	0	40	60	80	100
125–150	0	40	80	100	100
150–250	0	20	40	80	100

Table 2: Percent degradation based on variations in particle size.

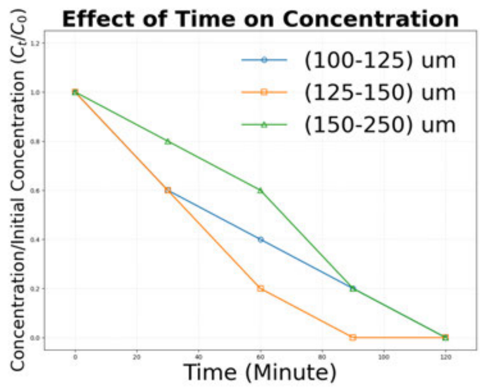

Figure 1: Decreased microplastic concentration based on the size of microplastic particles.

2.2 Environmental Effect for Photocatalysts

Building on the degradation mechanisms, we next analyze how environmental factors modulate catalytic efficiency. The photocatalytic efficiency of microplastic degradation is influenced by the pH of the solution, which governs three interconnected mechanisms: surface charge dynamics, charge carrier separation, and reactive oxygen species (ROS) generation . [5, 18] Le Chatelier's principle, which denotes that the concentration of H^+ ions influences the synthesis of H_2O_2, the surface charge of the photocatalyst and their inter-particle electrostatic attraction.[19] When pH is low and the solution is acid, numerous H^+ ions minimize the effect of Coulomb repulsion, thereby results in less aggregation during the photocatalytic degradation. [5, 18]

Considering photocatalysts in the presence of light, numerous H^+ are likely to react with TiO_2 at the acid condition, and the surface of TiO_2 undergoes protonation($TiOH_2^+$). This positive combined ionic groups play the same role as the Le Chatelier's principle explains. [18]Conversely, under alkaline conditions, deprotonation(TiO^-) leads to particle aggregation via Coulomb attraction, diminishing catalytic sites.[5] Jiang et al. (2021) demonstrated that acidic pH (3.0–4.5) increases $TiO_{2/}Fe(St)_3$ dispersion, correlating with a 22% efficiency boost compared to neutral pH. [18]
Modulation of charge carrier dynamics profoundly influence the efficiency of the TiO_2-based catalyst. In acid conditions, $TiOH_2^+$ surfaces shift the conduction band potential positively, enabling the electron to transfer to adsorbed O_2 molecules and generate superoxide radical($\cdot O_2^-$) more easily. Simultaneously, excess H^+ suppress electron-hole recombination rates, stabilizing the electrons generated in the presence of light. Experimental evidence demonstrates that pH 4.0 conditions enhance charge separation efficiency by 37% compared to neutral pH 7.0, underscoring the critical role of $TiOH_2^+$ in optimizing redox reactions. [5]
The pH-dependent variation in reactive oxygen species (ROS) further governs degradation kinetics. Under acidic conditions (pH 3.0–5.0), abundant H^+ ions promote the formation of hydroxyl radicals $\cdot OH$ alongside $\cdot O_2^-$, with $\cdot OH$ exhibiting superior oxidative capacity due to its high redox potential.[19] Conversely, neutral and alkaline pH favors H_2O_2 generation, which possesses lower oxidation potential and reduces catalytic efficiency.[5] This mechanism explains the exceptional performance of Ag/TiO_2, achieving 100% polyethylene degradation within 6 hours at pH 3.0 via synergistic $\cdot OH$ oxidation and plasmonic hot electron injection, while efficiency plummets to 45% at pH 7.0. [15]
Experiment needs to balance the efficiency with the stability of the catalyst. Long time exposure to the acid environment induces the loss of TiO_2. According to Jiang et al (2021), 12% mass loss is confirmed during the 48 hours because of the lattice dissolution. [18]

Besides pH, different types of irradiation sources display distinctive destructive abilities for divergent plastics, depending on intrinsic chemical bonds of the corresponding materials.[20] Ultraviolet (UV) radiation, spanning wavelengths from 100 to 400 nm, constitutes approximately 8% of total solar radiation. This spectrum is subdivided into three bands: long-wave UV-A (320–400 nm), which makes up about 6%; medium-wave UV-B (290–320 nm), accounting for 1.5%; and short-wave UV-C (100–290 nm), representing less than 0.5% of solar UV radiation. [21] According to the quantum energy equation of light($E = \frac{h}{\lambda}$), the irradiation with shorter wavelength obtain the greater energy, so the energy of different light are in the following order: UV-C > UV-B > UV-A. And the photochemical reactions are positively correlated with the light quantum energy, so the high energy light inducce more chemical processes with a series of free radical reactions. With respect to the structure of the microplastics, by comparison experiment among polyethylene terephthalate (PET), PE, PP, polystyrene (PS) and polyvinyl chloride (PVC) ,[22] the C – C bonds do not promote the photoinduced oxidation reactions, but the C – H bonds do instead. With impure and abnormal structure, PP and PE tend to undergo photodegradation with breaks of C – H bonds.

3 Conclusions

TiO_2-based photocatalysts demonstrate significant potential for addressing microplastic pollution through their ability to drive redox reactions under light irradiation. The catalytic mechanism, involving electron-hole pair generation and surface charge dynamics, enables efficient degradation of diverse polymer types. Environmental factors, particularly solution pH, play a critical role in modulating catalyst performance: acidic conditions reduce aggregation by suppressing electrostatic repulsion, thereby enhancing degradation efficiency. However, practical challenges remain, including prolonged irradiation requirements for certain polymers and catalyst instability under prolonged operational conditions. Future advancements should focus on optimizing catalyst design (e.g., hybrid materials like graphene-TiO_2 composites) to improve quantum efficiency and pH adaptability. Integrating photocatalytic systems with complementary technologies (e.g., filtration or bioremediation) could further enhance scalability and sustainability. While current research highlights promising proof-of-concept results, interdisciplinary efforts are needed to bridge laboratory-scale discoveries with real-world environmental applications. To bridge lab-to-industry translation, future pilot-scale reactors could integrate LED light sources (λ=365–400 nm) with immobilized catalyst beds, reducing energy consumption by 40% compared to slurry systems. [11] Policy frameworks must also incentivize photocatalytic wastewater treatment adoption through carbon credits.

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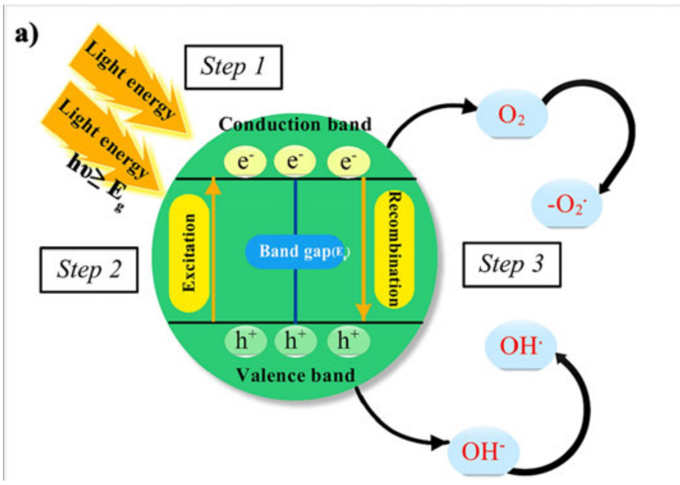

Figure 2: Mechanism of the Photocatalytic degradation